Generalized Bayes: the route back to standard Bayes I

The inference engine FedGVI federates is the generalized (Gibbs) posterior (Bissiri et al. 2016; Jiang and Tanner 2008; Jewson et al. 2018; Knoblauch et al. 2022), which trades the likelihood for a loss L and the KL regularizer for a general divergence $\mathcal{D}$:

$$q_n^*(s) = \arg\min_{q_n} \mathbb{E}_{q_n}\Big[\sum_i L(s; o_i)\Big] + \frac{1}{\tau}\,\mathcal{D}(q_n \,\|\, \pi_0), \tag{1}$$

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with prior π_0, learning rate τ, and regularizing divergence $\mathcal{D}$. The learning rate is part of the inferential specification, not a cosmetic constant; coarsened-posterior and safe-Bayes work show why calibration of that temperature matters under misspecification (Miller and Dunson 2018; Grünwald 2012), where ordinary Bayes concentrates around a KL pseudo-truth rather than literal truth when the model family is wrong (Kleijn and Vaart 2012). We name the object eq. 1 defines.

Definition (Generalized-(Gibbs)-Bayes posterior)

For a loss L, prior π_0, learning rate $\tau > 0$, and divergence $\mathcal{D}$, the generalized-Bayes posterior is the minimizer q_n^* of (1). For $\mathcal{D} = \mathrm{KL}$ the minimizer is the tempered softmax

$$q_n^*(s) \;\propto\; \pi_0(s)\,\exp\big(-\tau \textstyle\sum_i L(s; o_i)\big),$$

implemented in `generalized_bayes.generalized_posterior`.

The tempered softmax of eq. 1, stated in the definition above, is not an approximation: it is the exact closed-form minimizer of eq. 1 when the regularizer is the KL divergence, because the categorical support is finite and the objective is strictly convex in q. The recovery to standard Bayes follows by choosing the loss. With $L = \mathrm{NLL}$, $\mathrm{NLL}(p, o) = -\log p(o)$, the exponential in that tempered softmax becomes a product of likelihoods and the minimizer is *exactly* standard Bayes; eq. 6 in sec. 5.8 states that corner, and Corollary 6 there pins it to the closed-form prior-times-likelihood product. The largest observed discrepancy between `generalized_posterior` in this regime and the analytic Bayes posterior is 5.55e-17, reported in sec. 7.1 — exact to machine precision (a maximum deviation of about one ULP), not merely close.

FedGVI computes each client update against a *cavity* rather than the full posterior, so a contributing agent does not double-count its own previous message. We name that operation.

Definition (Cavity / PVI factor update)

The cavity removes agent n's factor from the colony posterior in natural-parameter (log) space,

$$q_{-n}(s) \;=\; \frac{q(s)/t_n(s)}{\sum_{s'} q(s')/t_n(s')} \;=\; \mathrm{softmax}(\,\log q(s) - \log t_n(s)),$$

where the final expression makes the normalization explicit; the partitioned-variational-inference (PVI) update re-multiplies a refreshed factor onto the cavity of (2). Taking a cavity and re-multiplying the original site factor restores the global posterior

$$q(s) = \frac{q_{-n}(s)t_n(s)}{\sum_{s'} q_{-n}(s')t_n(s')}\,, \tag{2}$$

with the original site factor, the recombination identity, the property `generalized_bayes.cavity` and `generalized_bayes.update_factor` satisfy.

The numbered recombination identity is eq. 2.

The cavity of eq. 2 is the discrete analogue of the expectation-propagation / partitioned-VI cavity used outside active inference (Ashman et al. 2022; Bui et al. 2018), imported here so that the per-agent generalized-Bayes update of eq. 1 is computed against the colony belief with the agent's own contribution removed — exactly the sensory-attenuation discipline the belief-sharing round of sec. 5.9 requires. What remains unspecified in eq. 1 are its two ingredients — the divergence $\mathcal{D}$ and the loss L — whose robust members and standard-Bayes limits sec. 5.6 develops next; the aggregation identity (sec. 5.8) then federates the resulting per-agent posteriors.

The authoritative notation supplement makes the same normalization and recombination contract explicit in eq. 30 and eq. 31; those equations govern the symbols used by the implementation and all later supplements.

Conjugate likelihood learning for the shared model I

Active-inference agents learn the parameters of their generative model, not just plan with them (Smith et al. 2022; Friston et al. 2024). The likelihood matrix A carries a Dirichlet prior with concentration a over each column, updated conjugately by accumulating observation-state co-occurrence counts (eq. 14), giving the column-normalized expected likelihood. The update of eq. 14 is driven by the expected sufficient statistics under the data-generating model, so as the concentrations accumulate $\mathbb{E}[A]$ converges to the true likelihood. Convergence is measured by the per-column KL divergence summed over hidden states, which decreases monotonically toward the standard-Bayes fixed point; sec. 7.3 reports the learning curve, where the KL falls from 3.4231 to 0.0027 across 24 count batches. A forgetting hyperprior optionally decays the running mass toward an asymptote so the agent does not become infinitely confident; with the hyperprior disabled the classical unbounded accumulation of eq. 14 is recovered. The implementation is `dirichlet_learning.learn_likelihood`.

Structure learning in the active-inference frame proceeds by Bayesian model reduction (BMR): given a full model with Dirichlet posterior `post` under prior `prior`, the change in negative variational free energy from swapping in a *reduced* prior — for example one that prunes a redundant column toward zero — is available in closed form without re-running inference (Friston and Penny 2011; Smith et al. 2022). Because the likelihood is shared, the reduced posterior is `post + reduced_prior - prior`, and the free-energy difference is a difference of log multivariate Beta functions (eq. 16), where $\ln B(a) = \sum_k \ln \Gamma(a_k) - \ln \Gamma(\sum_k a_k)$ is the log Dirichlet normalizer. A positive ΔF in eq. 16 means the reduced model has more evidence — the pruned structure was redundant and should be adopted; a negative ΔF means the reduction destroyed something the data support. When the reduced prior equals the prior the score is identically zero, the no-reduction fixed point. sec. 7.4 reports $\Delta F = 3.68$ for a redundant reduction (accepted) against $\Delta F = -27.67$ for a supported one (rejected). The implementation is `bayesian_model_reduction.reduce`.